PHYSICS

1) A particle is moving towards East with a velocity 10 m/s. In 10 seconds the velocity changes to 10 m/s. Northwards. The average acceleration during this period is :-

(1) $\sqrt{2}$ m/sec^2 along North-East direction

(2) $\sqrt{2}$ m/sec^2 along North-West direction

(3) $\frac{1}{\sqrt{2}}$ m/sec^2 along North-East direction

(4) $\frac{1}{\sqrt{2}}$ m/sec^2 along North-West direction

2) A particle moves in east with velocity of 15 m/s for 2 sec. then moves northward with 5 m/s for 8 sec, then average velocity of the particle is :-

(1) 1 m/s

(2) 5 m/s

(3) 7 m/s

(4) 10 m/s

3) A person walks 80 m east, then turns right through angle 143° walks further 50 m and stops. His position relative to the starting point is

(1) 50 m, 53° east of south

(2) 50 m, 53° south of east

(3) 30 m, 37° south of east

(4) 30 m, 53° south of east

4) Let $\vec{v}$ and $\vec{a}$ denote the velocity and acceleration respectively of a body in one-dimensional motion

(1) $|\vec{v}|$ must decrease when $\vec{a} < 0$

(2) speed must increase when $\vec{a} > 0$

(3) speed will increase when both $\vec{v} < 0$ and $\vec{a} > 0$

(4) speed will decrease when $\vec{v} < 0$ and $\vec{a} > 0$

5) If a car covers $\left(\frac{2}{5}\right)^{th}$ of the total distance with speed v_1 and $\left(\frac{3}{5}\right)^{th}$ distance with speed v_2 then average speed is

(1) $\sqrt{v_1 v_2}$

(2) $\frac{v_1 v_2}{2}$

(3) $\frac{5v_1 v_2}{2v_1 + 3v_2}$

(4) $\frac{5v_1 v_2}{3v_1 + 2v_2}$

6) A point traversed half a circle of radius r during a time interval t_0, its mean speed and magnitude of mean velocity are :-

(1) $\frac{\pi r}{t_0}, \frac{r}{t_0}$

(2) $\frac{2\pi r}{t_0}, \frac{2r}{t_0}$

(3) $\frac{\pi r}{t_0}, \frac{2r}{t_0}$

(4) $\frac{2\pi r}{t_0}, \frac{r}{t_0}$

7) A particle moves in a straight line with an acceleration a ms^{-2} at time 't' seconds where $a = -\frac{1}{t^2}$. When t = 1 the particle has a velocity of $3ms^{-1}$ then find the velocity when t = 4 :

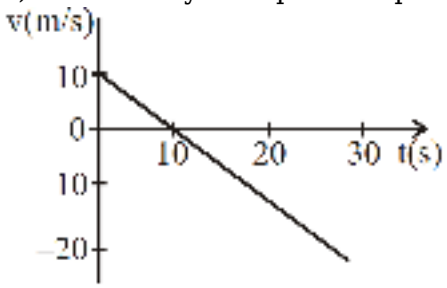
(1) 2.90 m/s

(2) 3.25 m/s

(3) 2.0 m/s

(4) 2.25 m/s

8) The velocity-time plot of a particle moving on a straight line shown in the figure



(1) The particle has a constant acceleration

(2) The particle has never turned around

(3) The particle has a zero displacement

(4) The average speed in the interval 0 to 10 s is the same as the average speed in the interval 10 s to 30 s

9) The engine of a motorcycle can produce a maximum acceleration 5 m/s^2. Its brakes can produce a maximum retardation 10 m/s^2.If motorcyclist start from point A and reach at point B. What is the minimum time in which it can cover if distance between A and B is 1.5 km :-
(Given : that motorcycle comes to rest at B)

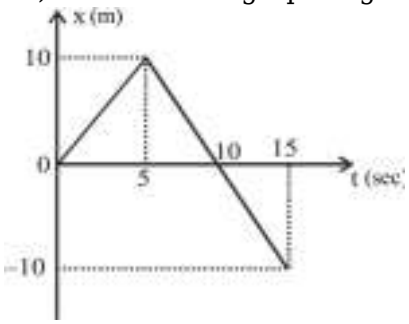
(1) 30 sec

(2) 15 sec

(3) 10 sec

(4) 5 sec

10) Position-time graph is given. Find average speed in time t = 0 and t = 15 sec :-



(1) 5 m/s

(2) $\frac{5}{3}$ m/s

(3) 2 m/s

(4) $\frac{2}{3}$ m/s

11) A particle released from a height covers $\frac{7}{16}$th part of total displacement in the last second of its journey. then its time of journey is :-

(1) 3 s

(2) 4 s

(3) 5 s

(4) 7 s

12) A body is moving towards north with initial velocity of 13 ms^{-1}. It is subjected to a retardation of 2 ms^{-2}. The distance travelled by it in 7th second is :-

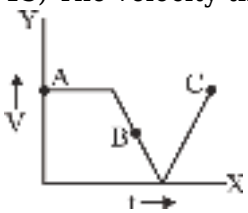
(1) Zero

(2) 0.75 m

(3) 0.25 m

(4) 0.5 m

13) The velocity time graph of a body is shown in Figure. It implies that at point B :-



(1) The acceleration is zero

(2) There is a acceleration towards motion

(3) There is a acceleration which opposes motion

(4) None of the above

14) Which of the following is always correct :-

(1) Magnitude of displacment is equal to the distance covered by particle

(2) Magnitude of average velocity is equal to the average speed

(3) Magnitude of instantaneous velocity will be equal to instantaneous speed

(4) None of these

15) A body starts from rest with uniform acceleration. What is the ratio of distance travelled during the 4th and 3rd sec.

(1) $\frac{7}{5}$

(2) $\frac{5}{7}$

(3) $\frac{7}{3}$

(4) $\frac{3}{7}$

16) The velocity v of a moving particle varies with displacement as $x = \sqrt{v+1}$. The acceleration of the particle at x = 5 unit will be :-

(1) $\sqrt{6}$ unit

(2) 24 unit

(3) 240 unit

(4) 25 unit

17) The displacement of a particle is given by $\sqrt{x} = 2t + 5$. What is the nature of the motion of the particle :-

(1) Uniformly accelerated

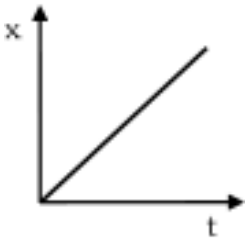
(2) Uniform motion

(3) Retarded

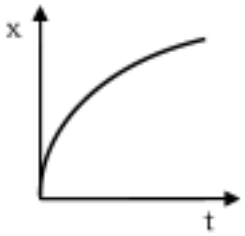
(4) Non uniformly acceerated

18) Which of the following position–time (x–t) graphs represents positive acceleration in rectilinear motion?

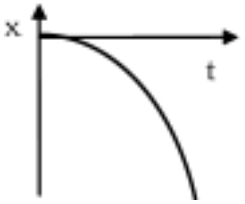
(1)



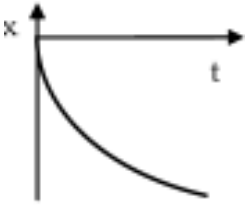
(2)



(3)



(4)



19) A body is projected vertically upwards with some velocity u. It is at the same point at t = 6 s and at t = 4s. Find the value of u :

(1) 30 m/s
(2) 40 m/s
(3) 20 m/s
(4) 50 m/s

20) A balloon is going upwards with velocity 12 m/sec. It releases a packet when it is at a height of 65 m from the ground. How much time the packet will take to reach the ground (g = 10 m/sec^2)

(1) 5 sec
(2) 6 sec
(3) 7 sec
(4) 8 sec

21) The displacement of an object along the three axes are given by $x = 2t^2$, $y = t^2 - 4t$ and $z = 3t - 5$. The initial velocity of the particle is :-

(1) 10 unit
(2) 12 unit
(3) 5 unit
(4) 2 unit

22) A particle is at rest initially find out the distance covered by particle in its 3rd sec if the acceleration of particle is 6 m/s^2 ?

(1) 12 m
(2) 10 m
(3) 15 m

(4) 20 m

23) A car is moving with a speed of 30 km/h, when the brakes are applied it retards uniformly and comes to rest after travelling 90 m distance, Now If same car moves with speed 50 km/h and if stopped by same retardation then distance travelled by it before it stops :-

(1) 360 m
(2) 180 m
(3) 250 m
(4) None

24) A particle is moving with uniform acceleration along a straight line . Its velocities at A & B are respectively 7 m/s & 17 m/s . M is mid point of AB . If t_1 is the time taken to go from A to M and t_2 the time taken to go from M to B, the ratio $\frac{t_1}{t_2}$ is equal to :-

(1) 3 : 2
(2) 3 : 1
(3) 2 : 1
(4) 2 : 3

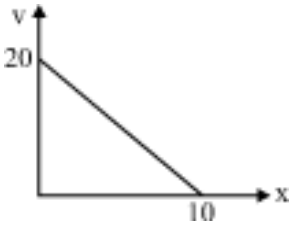
25) From the top of a tower, a particle is thrown vertically downwards with a velocity of 10 m/s. The ratio of the distances covered by it in the 3rd and 2nd second of the motion is (Take g = 10 m/s^2) :

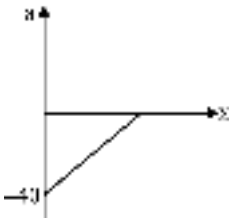
(1) 5 : 7
(2) 7 : 5
(3) 3 : 5
(4) 5 : 3

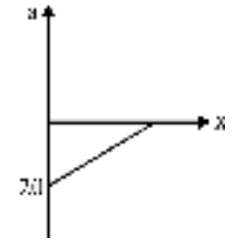
26) If a ball is thrown from a 20 m high building in downward direction with some velocity. If it takes 0.5 sec to strike the ground then find out time to strike ground if it is thrown in upward direction from same speed :-

(1) 4 sec
(2) 8 sec
(3) 0.5 sec
(4) 1 sec

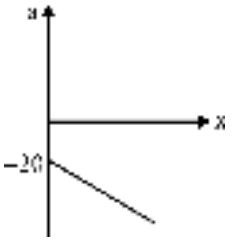
27) The given graph shows the variation of velocity with displacement. Which one of the graph given below correctly represents the variation of acceleration with displacement.



(1) 

(2) 

(3) 

(4) 

28) If velocity $v = 6t^2 + 2t + 3$ m/sec. then find average velocity from t = 0 to t = 1 sec.

(1) 11 m/sec.

(2) 8 m/sec.

(3) 6 m/sec.

(4) 7 m/sec.

29) A particle moves along the curve $y = \frac{x^2}{2}$. Here x varies with time as $x = \frac{t^2}{2}$. Where x and y are measured in metre and t in second. At t = 2s, the velocity of the particle (in ms^{-1}) is :

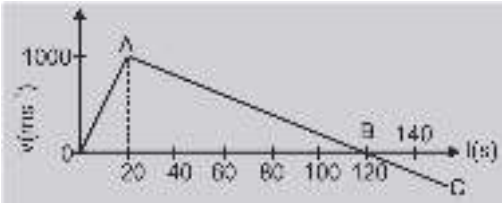
(1) $2\hat{i} - 4\hat{j}$

(2) $2\hat{i} + 4\hat{j}$

(3) $4\hat{i} + 2\hat{j}$

(4) $4\hat{i} - 2\hat{j}$

30) A rocket is launched upward from the earth surface, whose velocity time graphs shown in figure. Then maximum height attained by the rocket is :



(1) 1 km

(2) 10 km

(3) 100 km

(4) 60 km

31) For motion of an object along the x-axis the velocity v depends on the displacement x as $v = 3x^2 - 2x$, then what is the acceleration at x = 2m.

(1) 48 ms^{-2}

(2) 80 ms^{-2}

(3) 18 ms^{-2}

(4) 10 ms^{-2}

32) Two cars P and Q start from a point at the same time in a straight line and their positions are represented by $x_p(t) = at + bt^2$ and $x_Q(t) = ft - t^2$. At what time do the cars have the same velocity?

(1) $\frac{a+f}{2(1+b)}$

(2) $\frac{f-a}{2(1+b)}$

(3) $\frac{a-f}{1+b}$

(4) $\frac{a+f}{2(b-1)}$

33) A particle moves with a speed v in a horizontal circular path. The change in its velocity for covering 60° will be :-

(1) $v\sqrt{2}$

(2) $\frac{v}{\sqrt{2}}$

(3) $v\sqrt{3}$

(4) v

34) The engine of a train which is moving with constant acceleration passes an electric pole with a velocity 'u' and the last compartment of the train crosses the same pole with a velocity v. Then find the velocity with which the mid-point of the train passes the pole.

(1) $\sqrt{\frac{u^2-v^2}{2}}$

(2) $\frac{u-v}{2}$

(3) $\sqrt{\frac{u^2+v^2}{2}}$

(4) $\frac{u+v}{2}$

35) A particle has initial velocity $(4\hat{i}+4\hat{j})$ m/s and an acceleration $-0.4\hat{i}$ m/s^2, at what time will its speed be 5 m/s.

(1) 12.5 sec

(2) 2.5 sec

(3) 7.5 sec

(4) 10 sec

36) A body thrown vertically upwards so as to reach its maximum height in t second. The total time from the time of projection to reach a point at half of its maximum height while returning (in second) is :

(1) $\sqrt{2}\,t$

(2) $\frac{t}{\sqrt{2}}$

(3) $t\left(1+\frac{1}{\sqrt{2}}\right)$

(4) $\frac{3t}{2}$

37) A body starts from rest and travels a distance S with uniform acceleration, then moves uniformly a distance 2S and finally comes to rest after moving further 3S under uniform retardation. The ratio of the average velocity to maximum velocity is :-

(1) 2/5

(2) 3/5

(3) 4/7

(4) 5/7

38) A particle moves along a straight line in such a way that it's acceleration in increasing at the rate of 2m/s^3. It's initial acceleration and velocity were 0, the distance covered by it in t = 3 second is :-

(1) 27 m

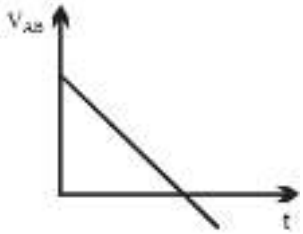
(2) 9 m

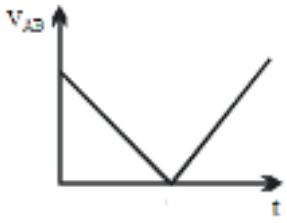
(3) 3 m

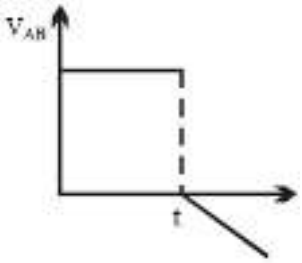
(4) 1 m

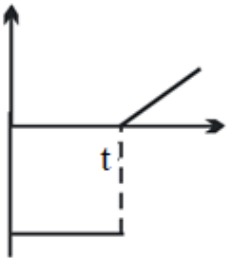
39) A body A is thrown vertically upwards with such a velocity that it reaches a maximum height of *h*. Simultaneously another body *B* is dropped from height *h*. It strikes the ground and does not rebound. The velocity of *A* relative to *B* v/s time graph is best represented by (upward direction is

positive)

(1) 

(2) 

(3) 

(4) 

40) From a balloon rising vertically upwards at 5m/s a stone is thrown up at 10m/s relative to the balloon. Its velocity with respect to ground after 2sec is (assume g = 10m/s^2)

(1) 0
(2) 20 m/s
(3) 10 m/s
(4) 5 m/s

41) A bus starts from rest with an acceleration of 1 m/sec^2. A man who is 48 m behind the bus is moving with a uniform velocity of 10 m/sec. Then the minimum time after which the man will catch the bus is :-

(1) 12 sec.
(2) 8 sec.
(3) 10 sec.
(4) 4.8 sec.

42) If $\vec{u} = \hat{i} \times (\vec{a} \times \hat{i}) + \hat{j} \times (\vec{a} \times \hat{j}) + \hat{k} \times (\vec{a} \times \hat{k})$ then :-

(1) $\vec{u}$ is a unit vector

(2) $\vec{u} = \vec{a} + \hat{i} + \hat{j} + \hat{k}$

(3) $\vec{u} = 2\vec{a}$

(4) $\vec{u} = 8(\hat{i} + \hat{j} + \hat{k})$

43) If $|\hat{A} \times \hat{B}| = \frac{3}{4}\hat{A}.\hat{B}$, then $|\hat{A} - \hat{B}| = ...$

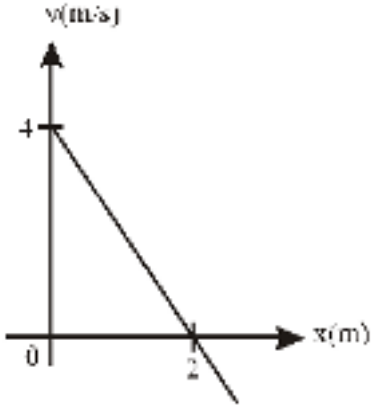
(1) $\sqrt{\frac{18}{5}}$

(2) $\sqrt{\frac{2}{5}}$

(3) $\sqrt{\frac{8}{5}}$

(4) $\sqrt{\frac{10}{5}}$

44) The velocity (v) of a particle moving along x-axis varies with its position x as shown in figure. The acceleration (a) of particle varies with position (x) as



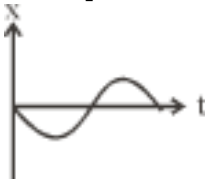
(1) $a^2 = x + 3$

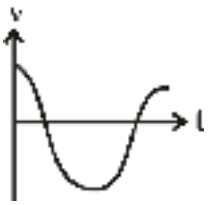
(2) $a = 2x^2 + 4$

(3) $2a = 3x + 5$

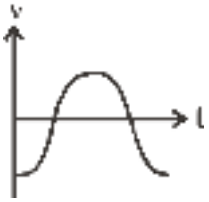
(4) $a = 4x - 8$

45) If position time graph of a particle is sine curve as shown what will be its velocity - time graph.



(1) 

(2) 

(3) 

(4) 

CHEMISTRY

1) 200 units of energy is required to take away electron from the lowest energy level to infinity. The energy released when electron falls from infinite level to the second shell (Assuming Bohr model to be applicable) will be –

(1) 100 units
(2) 400 units
(3) 50 units
(4) 25 units

2)

If the mass of e is doubled and charge is halved. The energy of e^- in 1st orbit of H-atom would be

(1) –13.6 ev
(2) –1.7 ev
(3) –0.85 ev
(4) –27.2 ev

3) When electron in Bohr atom is excited then which of the following increases

(1) Potential energy
(2) Time period of revolution
(3) Angular momentum
(4) All of these

4)

If two neutrons are added to an element, then it will get converted to its :-

(1) Isotope

(2) Isotone

(3) Isobar

(4) None of these above

5) Increasing order of e/m ratio is :-

(1) e^-, p, n, α

(2) n, p, e^-, α

(3) n, p, α, e^-

(4) n, α, p, e^-

6) Difference in wavelength of two extreme spectral lines of H-atom in Balmer region is: (Where R_H is Rydberg constant)

(1) 7.2 / R_H

(2) 0.25 / R_H

(3) 4 / R_H

(4) 3.2 / R_H

7) Species which are isoelectronic to one another are

a	b	c	d	e
CN^-	OH^-	O_2	N_2	CO

Correct Ans :-

(1) a, b, c

(2) a, c, d

(3) a, d, e

(4) b, c, d

8) The nucleus of an atom can be assumed to be spherical. The radius of the nucleus of mass number A is given by $1.25 \times 10^{-13} \times A^{1/3}$ cm. Radius of atom is one Å. If the mass number is 64, then the fraction of the atomic volume that is occupied by the nucleus is

(1) 1.0×10^{-3}

(2) 5.0×10^{-5}

(3) 2.5×10^{-2}

(4) 1.25×10^{-13}

9) If mass of electron become half, mass of proton become double, and mass of neutron become 3/4th than new mass of $_8O^{16}$ will be :-

(1) 37.5 % increase

(2) Remain unchange

(3) 12.5 % increase

(4) 25 % decrease

10) A certain particle carries 1×10^{-16} C of electric charge. Calculate the number of electrons present in :-

(1) 100

(2) 1000

(3) 1.6×10^{-19}

(4) 9.1×10^{-31}

11) In a hydrogen atom, the electron is at a distance of 4.768 Å from the nucleus. The angular momentum of the electron is :-

(1) $\frac{3h}{2\pi}$

(2) $\frac{h}{2\pi}$

(3) $\frac{h}{\pi}$

(4) $\frac{2h}{\pi}$

12)

The ratio of area of orbit of first excited state of electron to the area of orbit of ground level, for hydrogen atom, will be :-

(1) 16 : 1

(2) 4 : 1

(3) 2 : 1

(4) None of these

13) The atom A, B, C have the configuration
A → [Z(90) + n(146)], B → [Z(92) + n(146)],
C → [Z(90) + n(148)] So that :-
(a) A and C - Isotones
(b) A and C - Isotopes
(c) A and B - Isobars
(d) B and C - Isobars
(e) B and C - Isotopes
The wrong statement's are:-

(1) a, b only

(2) c, d, e only

(3) a, c, d only

(4) a, d, e only

14) A divalent cation is isoelectronic of CO_2 and has (Z + 2) neutrons. What is ionic mass of divalent cation :-

(1) 24

(2) 26

(3) 50

(4) 48

15) In H-atom, electron transits from 6th Orbit to 2nd orbit in multi step. Then total spectral lines (without Balmer series) will be :-

(1) 6

(2) 10

(3) 4

(4) 0

16)

Maximum frequency of emission is obtained for the transition :–

(1) n = 1 to n = 2

(2) n = 2 to n = 6

(3) n = 2 to n = 1

(4) n = 6 to n = 2

17) The ratio of the difference in energy between the first and the second Bohr orbit to that between the second and the third Bohr orbit of H is -

(1) $\frac{1}{2}$

(2) $\frac{1}{3}$

(3) $\frac{4}{9}$

(4) $\frac{27}{5}$

18) The radii of two of the first four Bohr's orbits of the hydrogen atom are in the ratio 1 : 4. The energy difference between them may be :-

(1) Either 12.09 eV or 3.4 eV

(2) Either 2.55 eV or 10.2 eV

(3) Either 13.6 eV or 3.4 eV

(4) Either 3.4 eV or 0.85 eV

19) Which is not true with respect to cathode rays ?

(1) A stream of electrons

(2) Charged particles

(3) Move with speed as that of light

(4) Can be deflected by magnetic fields

20) In gold foil experiment number of deflected α-particles at angle 90° is 63, then number of α-particle deflected at 120° is:

(1) 112

(2) 42

(3) 56

(4) 28

21) In an atom, two electrons move around the nucleus in circular orbits of radii R and 4R, the ratio of the time taken by them to complete one revolution is (Considering Bohr's atomic theory is applicable)-

(1) 1 : 4

(2) 4 : 1

(3) 1 : 8

(4) 8 : 1

22) Match the following :-

(A)	Energy of ground state of He^+	(i)	+ 6.04 eV
(B)	Potential energy of I orbit of H–atom	(ii)	– 27.2 eV
(C)	Kinetic energy of II excited state of He^+	(iii)	8.72×10^{-18} J
(D)	Ionisation potential	(iv)	– 54.4 eV

(1) A – (i), B – (ii), C – (iii), D – (iv)

(2) A – (iv), B – (iii), C – (ii), D – (i)

(3) A – (iv), B – (ii), C – (i), D – (iii)

(4) A – (ii), B – (iii), C – (i), D – (iv)

23) The angular momentum of an electron in a given orbit is J, its kinetic energy will be :

(1) $\frac{1}{2}\frac{J^2}{mr^2}$

(2) $\frac{Jv}{r}$

(3) $\frac{J^2}{2m}$

(4) $\frac{J^2}{2\pi}$

24) Choose the Incorrect relation on the basis of Bohr's theory -

(1) velocity of electron $\propto \frac{1}{n}$

(2) frequency of revolution $\propto \frac{Z^2}{n^3}$

(3) radius of orbit $\propto n^2Z$

(4) force on electron $\propto \frac{Z^2}{n^4}$

25) **Assertion (A) :-** All isotopes of a given element show the same type of chemical behaviour.
Reason (R) :- The chemical properties of an atom are controlled by the number of electron in the atom.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is incorrect but (R) is correct

(4) Both (A) and (R) are correct but (R) is the correct explanation of (A)

26) **Assertion :** Most of the α-particle striking gold foil in Rutherford's experiment were deflected.
Reason : The size of nucleus is very large as compared to the size of atom.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

27) **Assertion :** Total energy of an electron in Hydrogen atom is negative.
Reason : Negative sign indicates electron is bounded towards nucleus.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

28) When electrons of hydrogen atom return to L-shell from the shells of higher energy, we get a series of line in the spectrum. This series is called

(1) Lyman series

(2) Balmer series

(3) Brackett series

(4) Paschen series

29)

Ratio of slopes of 'KE_{max}' v/s 'ν' and 'V_0' vs 'ν' curves in photoelectric effect gives
(ν is frequency of incident radiation & V_0 is stopping potential)

(1) Charge of electron (e)

(2) Planck's constant (h)

(3) Work function (w)

(4) $\frac{h}{e}$

30) At the room temperature if all the ultravoilet to infrared wavelength's radiations pass into the hydrogen gas then absorbed lines are :-

(1) Lyman, Balmer, Paschan

(2) Lyman, Paschen

(3) Lyman series

(4) Balmer series

31)

Kinetic energy of photoelectrons is independent on _____ of incident radiation

(1) Wavelength

(2) Wave number

(3) Frequency

(4) Intensity

32) If the shortest wavelength of H atom in Lyman series is x. then longest wavelength in Balmer series of He^+ is

(1) $\frac{9x}{5}$

(2) $\frac{36x}{5}$

(3) $\frac{x}{4}$

(4) $\frac{5x}{9}$

33) For which transition in H-atom, the amount of energy released will be maximum?

(1) n = 4 to n = 2

(2) n = 5 to n = 2

(3) n = 2 to n = 1

(4) n = 7 to n = 2

34) Let v_1 be the frequency of the series limit of the Lymen series, v_2 be the frequency of the first line of the lymen series and v_3 be the frequency of the series limit of Balmer series :-

(1) $v_1 - v_2 = v_3$

(2) $v_2 - v_1 = v_3$

(3) $v_2 = \frac{1}{2}(v_1 - v_3)$

(4) $v_1 + v_2 = v_3$

35) When a metal surface is irradiated by light of wavelength 300 nm; the stopping potential is found to be 0.5V. The work function of metal is:

(1) 2.4 eV
(2) 3.6 eV
(3) 6.3 eV
(4) 5.4 eV

36)

Which of the following statement is true in the context of photoelectric effect ?

(1) The kinetic energy of ejected electron is independent of the intensity of a radiation.
(2) The number of photoelections ejected depends upon the intensity of the incident radiation.
(3) The kinetic energy of the emitted electrons depends on the frequency of the incident radiation.
(4) All of these

37)

Threshold frequency of metal is ν_0. When light of frequency $\nu = 2\nu_0$ is incident on the metal plate, velocity of electron emitted is v_1 when a frequency of incident radiation is 5 ν_0, v_2 is velocity of emitted electron then $v_1 : v_2$

(1) 1 : 4
(2) 1 : 2
(3) 2 : 1
(4) 4 : 1

38)

Number of possible spectral lines which may be emitted in bracket series in H atom if electron present in 9th excited level returns to ground level, are

(1) 21
(2) 6
(3) 14
(4) 7

39)

The line spectra of two elements are not identical because

(1) The elements do not have the same number of neutrons
(2) They have different mass numbers
(3) Their outermost electrons are at different energy levels

(4) They have different valencies

40) The first Lyman transition in the hydrogen spectrum has ΔE = 10.2 eV. The same energy change is observed in the second Balmer transition of :-

(1) Li^{2+}

(2) Li^{+}

(3) He^{+}

(4) Be^{3+}

41) Radiations of frequency 'v' are incident on a photosensitive metal. The maximum kinetic energy of photoelectrons is 'x'. When the frequency of incident radiation is doubled, the maximum kinetic energy becomes ....

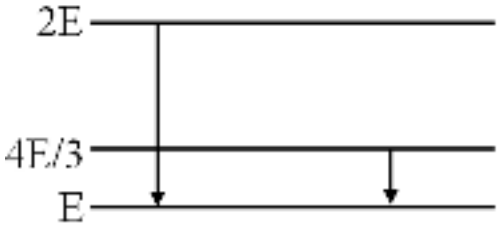
(1) 2x

(2) $\frac{x}{2}$

(3) > 2x

(4) $< \frac{x}{2}$

42) The given diagram indicates the energy levels of a certain atom. When the system moves from 2E level lo E level, a photon of wavelength l is emitted. The wavelength of the photon emitted during its transition from 4E/3 level to E level is.



(1) λ/3

(2) 3λ/4

(3) 4λ/3

(4) 3λ

43) An excited state of H atom emits a photon of wavelength λ and returns in the ground state, the principal quantum number of excited state is given by :-

(1) $\sqrt{\lambda R(\lambda R-1)}$

(2) $\sqrt{\frac{\lambda R}{(\lambda R-1)}}$

(3) $\frac{1}{\sqrt{\lambda R(\lambda R-1)}}$

(4) $\sqrt{\frac{(\lambda R-1)}{\lambda R}}$

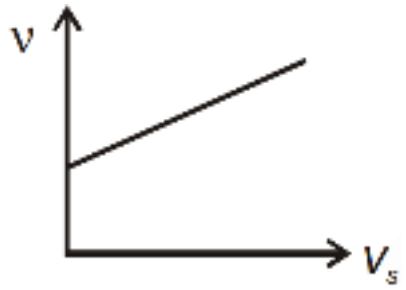
44) Radiation corresponding to the transition n = 4 to n = 2 in hydrogen atoms falls on a certain

metal (work function = 2.5 eV). The maximum kinetic energy of the photo-electrons will be :

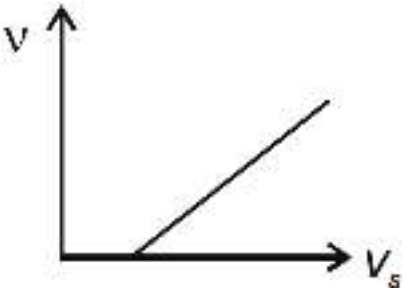
(1) 0.55 eV
(2) 2.55 eV
(3) 4.45 eV
(4) None of these

45) Graph of incident frequency with stopping potential in photoelectric effect is

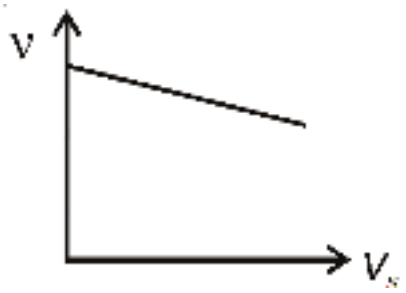
(1)



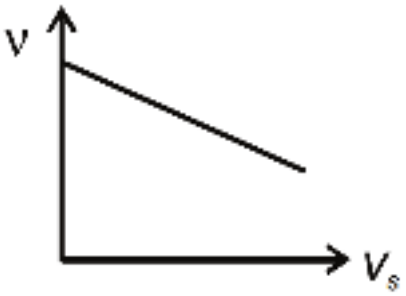
(2)



(3)



(4)



BIOLOGY

1) In centriole, total microtubules__A__and radial spoke__B__.

	(A)	(B)
(1)	18	9
(2)	27	9

(3)	18	27
(4)	9	20

(1) 1
(2) 2
(3) 3
(4) 4

2) Some of the enzymes, which are associated in converting fats into carbohydrates, are present in:

(1) Liposomes
(2) Golgi bodies
(3) Microsomes
(4) Glyoxysomes

3) Highest arm ratio occur in which chromosome :-

(1) Submetacentric
(2) Acrocentric
(3) Metacentric
(4) Telocentric

4) An enzymatic protein in flagella having ability of hydrolysis of ATP, is :-

(1) Actin
(2) Dynein
(3) Tubulin
(4) Nexin

5) Sometimes a few chromosome have non-staining secondary constriction at a constant location. This gives the appearance of a small fragment beyond the secondary constriction called :-

(1) Centriole
(2) Centromere
(3) Satellite
(4) Both (2) and (3)

6) Consider the following four statements (A - D) and select the option which includes all the correct ones only:
(A) The interphase nucleus has highly extended and elaborate nucleoprotein fibres called centromere.
(B) The inner membrane of nucleus usually remains continuous with the endoplasmic reticulum.
(C) The nuclear matrix contains nucleolus and microsome.
(D) The nucleoli are spherical structures present in the nucleoplasm.

(1) Statement A, B & D
(2) Statement A & D

(3) Statement D only
(4) Statement C & D

7) What is Axoneme in Cilia?

(1) The core of cilia
(2) Peripheral tubuler
(3) Radial spoke
(4) Outer membrane

8) Which one is **not** geneticaly less active structure?

(1) Euchromatin
(2) Heterochromatin
(3) NOR
(4) Nucleolus

9) The core of a flagellum composed of microtubules and their associated proteins is called :-

(1) Basal Body
(2) Microbodies
(3) Axoneme
(4) Microfilaments

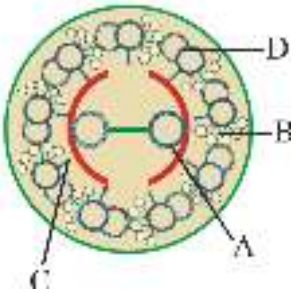
10) A common characteristic feature of plant sieve tube cell and most of mature mammalian erythrocytes is:-

(1) Absence plasma membrane
(2) Presence of Cell wall
(3) Presence of Haemoglobin
(4) Absence of Nucleus

11) One genome is which type of set of chromosomes?

(1) Haploid
(2) Diploid
(3) Triploid
(4) Polyploid

12) Given below is the diagrammatic sketch of a section of flagella. Identify the parts labelled A, B, C



and D and select the right option about them :-

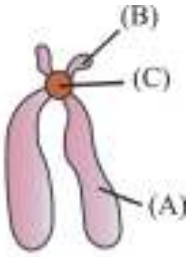
	Part-A	Part-B	Part-C	Part-D
(1)	Peripheral microtubule	Radial spoke	Interdoublet bridge	Central microtubule
(2)	Peripheral microtubule	Interdoublet bridge	Radial spoke	Central microtubule
(3)	Central microtubule	Radial spoke	Interdoublet bridge	Peripheral microtubule
(4)	Central microtubule	Interdoublet bridge	Radial spoke	Peripheral microtubule

(1) 1

(2) 2

(3) 3

(4) 4



13) Identify A, B and C in the given diagram below :-

	(A)	(B)	(C)
(1)	Short Arm	Long Arm	Centromere
(2)	Long Arm	Short Arm	Centrosome
(3)	Long Arm	Short Arm	Primary Constriction
(4)	Long Arm	Satellite	Centriol

(1) 1

(2) 2

(3) 3

(4) 4

14) Function of centrosome is:-

(1) Secretion

(2) Respiration

(3) Water regulation

(4) Arrangement of spindle fibres

15)

Select incorrect statement for ribosome

(1) These are smallest cell organelle
(2) These are universal cell organelle
(3) Only eukaryotic cytoplasm have ribosomes
(4) Ribosomes are membrane less cell organelle

16)

Which of the following enzyme degrade peroxide(H_2O_2)?

(1) Cathepsin
(2) Protease
(3) ATPase
(4) Catalase

17)

Organelle within organelle is characterised by all except-

(1) Non-membranous
(2) site of protein synthesis
(3) made up of two similar size units
(4) also freely present in cytoplasm

18)

Which r-RNA common between 70s & 80s ribosome

(1) 16s
(2) 18s
(3) 5s
(4) 58s

19)

Many microtubules are running parallely to the

(1) Radial axis of cillia
(2) Longitudinal axis of flagella
(3) Transverse axis of flagella
(4) Only in core of cillia

20) Read the following statements about cillium or flagellum and find out the **correct** statements.
a. Prokaryotic and eukaryotic flagella are structurally similar.
b. Cilia work like oars and comparatively smaller than flagella.

c. The core of cilium or flagellum is called axoneme.
d. Central tubules are connected by interdoublet bridges and is enclosed by central sheath.
e. Central sheath is connected to one of the tubules of each peripheral doublets by a radial spoke, There are nine radial spokes.
f. Peripheral doublets are also interconnected by A-B linkers [Nexin protein] .

(1) b, d, e, f
(2) c, d, e
(3) a, b, c
(4) b, c, e, f

21) **Statement A :** Content of nucleolus remains continuous with rest of cytoplasm.
Statement B : Large and numerous nucleoli are present in actively protein synthesizing cell.

(1) Statement A is true while statement B is false
(2) Both statements A & B are true
(3) Statement A is false while statement B is true
(4) Both statements A and B are false

22)

Centrosome is absent in ____

(1) Cell of higher plants
(2) cell of lower animal
(3) cell of higher animals
(4) cells of lower plants

23) Site of rRNA synthesis is :-

(1) Cytoplasm
(2) Nucleus
(3) Nucleolus
(4) ER

24) Which of the following organelle contains enzyme for β-oxidation?

(1) Glyoxysome
(2) Centrosome
(3) Peroxisome
(4) Sphaerosome

25) Choose the appropriate answer regards acrocentric chromosomes.

(1) The centromeres is situated close to its end, forming one extremely short and one very long arm
(2) The centromeres is situated at its end forming two extremely short and one very long arm.
(3) Chromosomes has a terminal centromere.

(4) Chromosomes has centromere slightly away from the middle

26) A student done the cell fractionation of a tissue and forgot to label his tubes. The content of one tube when studied showed organelles bounded by membrane with activity of nucleases enzyme (acid hydrolases). The organelles could be

(1) SER

(2) Chloroplast

(3) Lysosome

(4) Peroxisome

27) Read the following statements (A–D)
(A) Nuclear matrix or nucleoplasm contains nucleolus and chromatin
(B) Outer nuclear membrane usually remains continuous with the ER and also bears ribosomes on it
(C) Nuclear pores are the passages through which movement of protein molecules take place in only one direction from nucleus to the cytoplasm.
(D) Nucleous act as ribosome factory
Which of the above statements are correct?

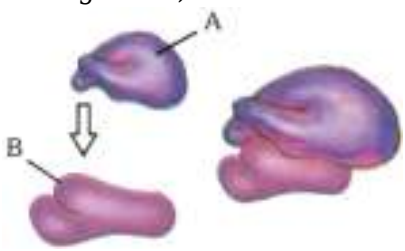
(1) A, B, C, D

(2) Only C

(3) A, C, D

(4) A, B, D

28) Here 'S' (Svedberg's unit) stands for sedimentation coefficient then what are [A] & [B] for



eukaryotic cell.

(1) A - 40S B-60S

(2) A - 60S B-40S

(3) A-80S B-60S

(4) A-40S B-80S

29) Select the *incorrect* match :

(1) Lampbrush chromosomes – Diplotene bivalents

(2) Allosomes – Sex chromosomes

(3) Submetacentric chromosomes – L-shaped chromosomes

(4) Polytene chromosomes – Oocytes of amphibians

30) The content of the nucleolus is continuous with the rest of the ..... as it is a ..... structure :

(1) Nucleoplasm, Double membranous

(2) Nucleoplasm, Membrane less

(3) Cytoplasm, Membrane less

(4) Cytoplasm, Single membranous

31) Incorrect statement in relation to nucleolus is :

(1) It is a spherical structure

(2) It is separated from nucleoplasm by nuclear envelop

(3) it is site of rRNA synthesis

(4) They are larger and more numerous in cells actively engaged in protein synthesis

32) Karyotheca or nuclear membrane is absent in:

(1) Nostoc

(2) Rhizobium

(3) Drosophila

(4) Both in 1 & 2

33) A cell organelle 'X' is divided into two types on the basis of a cell organelle 'Y', that helps in the protein synthesis. Identify 'X' and 'Y' respectively

(1) Golgi complex and ribosome

(2) ER and mitochondria

(3) Lysosome and ER

(4) ER and ribosome

34) Space between the two membrane of nucleus is :

(1) 10 - 50 nm

(2) 10 - 15 Å

(3) 75 Å

(4) 6 - 10 nm

35) **Assertion:** Erythrocytes of many mammals and sieve tube of vascular plants lack nucleus.
Reason: Some living Eukaryotic cells lack nucleus.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

36) **Assertion** : Centrioles helps in cell division.
Reason : Centrioles are absent in almost all plant cells.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but the Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

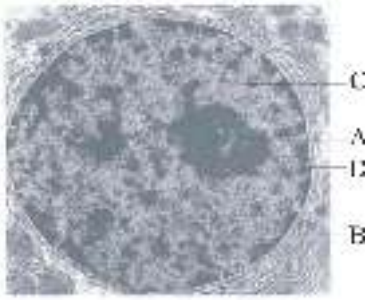
37) **Assertion:** Position of centromere decides the types of chromosome.
Reason: Every chromosome essentially has a primary constriction or the centromere.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.



38) Recognise the figure and find out the correct matching :-

	A	B	C	D
1	Nucleolus	Nucleoplasm	Nuclear pore	Nuclear membrane
2	Nucleoplasm	Nuclear membrane	Nucleolus	Nuclear pore
3	Nuclear pore	Nucleoplasm	Nucleolus	Nuclear membrane
4	Nucleolus	Nuclear membrane	Nucleoplasm	Nuclear pore

(1) 1

(2) 2

(3) 3

(4) 4

39) Given below are two statement -
Statement-I :- At number of places the nuclear envelope is interrupted by minute pores.
Statement-II :- Material of the nucleus stained by the acidic dye was given the name chromatin by Flemming.
In the light of the above statements, choose the correct answer from the options given below -

(1) Both statement I and statement II are correct

(2) Both statement I & statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

40) **Statement-I :** The interphase nucleus has highly extended and elaborate nucleoprotein fibres called chromatin.
Statement-II : Chromatin contains DNA and some histone proteins but lack non-histone proteins and RNA.

(1) Statement-I and II both are correct

(2) Statement-I and II both are incorrect

(3) Only Statement-I is correct

(4) Only Statement-II is correct

41) Read both statements and select the correct option.
Statement–I: Chromosomes are condensed structures formed during cell division.
Statement–II: Kinetochores are sites of attachment of spindle fibers.

(1) Both Statements are correct.

(2) Statement-I is correct Statement-II is incorrect.

(3) Statement-I is incorrect Statement-II is correct.

(4) Both Statements are incorrect.

42) Read the following statement and choose the correct option.
Statement–I :- Every chromosome essentially has a secondary constriction.
Statement–II :- A few chromosomes have non staining secondary constriction at a constant location.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

43) Which is correct option amongst the following statements ? a. Nuclear membrane, chloroplasts, mitochondria, microtubules and pili are absent in prokaryotic cells.
b. Nuclear membrane, chloroplasts, mitochondria, microtubules and pili are present in eukaryotic cells. c. Ribosomes are 70S in prokaryotic cells, chloroplasts and mitochondria. They are 80S in animal cells.

(1) a and b are wrong, c is correct

(2) a is correct, b and c are wrong

(3) a and b are correct, c is wrong

(4) a and c are correct, b is wrong

44)

In which of the following structure, ribosome is absent

(1) mitochondria

(2) chloroplast

(3) leucocyte

(4) viruses

45) Each chromosomes essentially has :

(1) Secondary constriction

(2) Centrioles

(3) Primary constriction

(4) Both (2) and (3)

46) Match the following :-

Column-I		Column-II	
a	Alkaloid	I	Vinblastin, curcumin
b	Essential oils	II	Morphine, Codeine
c	Toxins	III	Lemon grass oil
d	Drugs	IV	Abrin, Ricin

(1) a-III, b-II, c-I, d-IV

(2) a-III, b-II, c-IV, d-I

(3) a-II, b-III, c-I, d-IV

(4) a-II, b-III, c-IV, d-I

47) **Assertion :** Tertiary structure of protein molecules makes them biologically active.
Reason : Tertiary configuration of protein molecules is maintained by a single type of bond only.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

48) Choose the mismatch pair :

(1) Glutamic acid – Acidic

(2) Lysine – Basic

(3) Valine – ⊕ve Charged

(4) Phenylalanine – Aromatic

49) Match the columns I and II, and choose the correct combination from the options given:-

	Column-I(Component)		**Column-II(% of the total cellular mass)**
(a)	Ions	(i)	1
(b)	Lipids	(ii)	2
(c)	Carbohydrates	(iii)	3
(d)	Nucleic acids	(iv)	5–7
(e)	Proteins	(v)	10–15

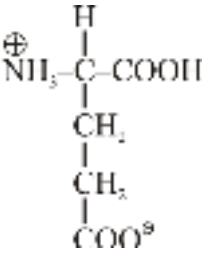
(1) a–i, b–ii, c–iii, d–iv, e–v

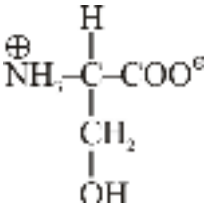
(2) a–ii, b–iii, c–i, d–v, e–iv

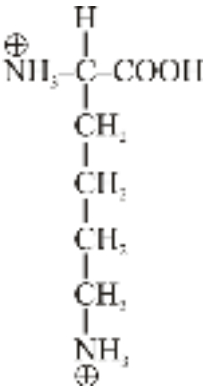
(3) a–iii, b–i, c–ii, d–iv, e–v

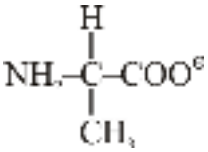
(4) a–iv, b–ii, c–iii, d–v, e–i

50) From the given below structure, identify the zwitter ions ?

(A) 

(B) 

(C) 

(D) 

(1) A and C
(2) Only A
(3) B and D
(4) A and B

51) Lipids are extracted with acid insoluble fraction because :-

(1) its molecular weight is less than 800 Da
(2) it form vesicle or micelle which are not water soluble
(3) it is polymer
(4) it is not a polymer

52) The acid insoluble fraction of the tissue has

(1) Amino acids, Proteins and nucleic acids
(2) Monosaccharides and polysaccharides
(3) Nucleic acids, nucleotides and nitrogen base
(4) Proteins, nucleic acids, polysaccharides and lipids

53) Which of the following element is present in negligible amount in human body :-

(1) N
(2) O
(3) Si
(4) H

54) Given below are list of some biomolecules
(a) Keratin (b) Nuclease

(c) Insulin (d) Glucose
(e) Glycine (f) Fat
How many of them are macromolecules and are present in retentate during analysis of biomolecules?

(1) Two
(2) Three
(3) Four
(4) Five

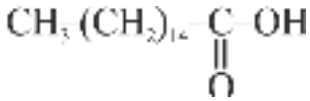
55) The amino acid phenylalanine is poorly soluble in water because its side chain or R group is

(1) Hydrophilic
(2) Heterocyclic chain
(3) Aromatic and non-polar
(4) Aromatic and polar

56) The most diverse chemical in a living being is :

(1) Polysaccharide
(2) Lipid
(3) Protein
(4) Nucleic acid

57) Given below is the structure of :-



(1) oleic acid
(2) Galactose
(3) Arachidonic acid
(4) Palmitic acid

58) Most abundant element in human body is :-

(1) Hydrogen
(2) Oxygen
(3) Nitrogen
(4) Carbon

59) In glycogen, which kind of linkage is found between adjacent glucose molecule?

(1) 1, 4
(2) 1, 6
(3) Both (1) & (2)
(4) 1, 2

60) Fats or oils are

(1) Monoglycerides
(2) Diglycerides
(3) Triglycerides
(4) None of the above

61) How many types of amino-acids are used to form protein?

(1) 18
(2) 20
(3) 21
(4) 22

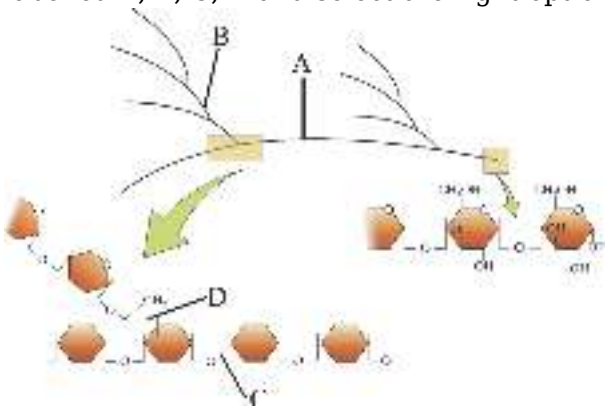
62) Which one bond is formed in between oxygen of amide and hydrogen of another amide in proteins :

(1) Hydrophobic bond
(2) Ionic bond
(3) Hydrogen bond
(4) Covalent bond

63) Specificity of different types of proteins is due to:-

(1) Types of amino acids
(2) Sequence of amino acids
(3) Number of amino acids
(4) Relative amino acid composition

64) Given below the diagrammatic sketch of a certain type of polysaccharide identify the parts labelled A, B, C, D and select the right option about them :-



	Part A	Part B	Part C	Part D
(1)	Linear chain	Branching chain	α(1–4) glycosidic linkage	α(1–6) glycosidic linkage
(2)	Branching chain	Linear chain	α(1–6) glycosidic linkage	α(1–4) glycosidic linkage

(3)	Branching chain	α(1–6) glycosidic linkage	α(1–4) glycosidic linkage	Linear chain
(4)	Linear chain	Branching chain	α(1–6) glycosidic linkage	α(1–4) glycosidic linkage

(1) 1

(2) 2

(3) 3

(4) 4

65) Amino acids are organic compounds containing an/a

(1) Amino group and an acidic group as substituents on the same carbon

(2) Amino group and an alcoholic group as substituents on the same carbon

(3) Hydroxyl group and a carboxyl group as substituents on the same carbon

(4) Hydroxyl group and a carboxyl group as substituents on adjacent carbons

66) Read the following four statements (A-D) :-
(A)The relative abundance of carbon and hydrogen with respect to other elements is higher in a living
organism than in earth's crust.
(B) α-amino acids are substituted ethane.
(C) Glycine is optically active.
(D)Gingely oil has lower melting point.
How many of the above statements are wrong ?

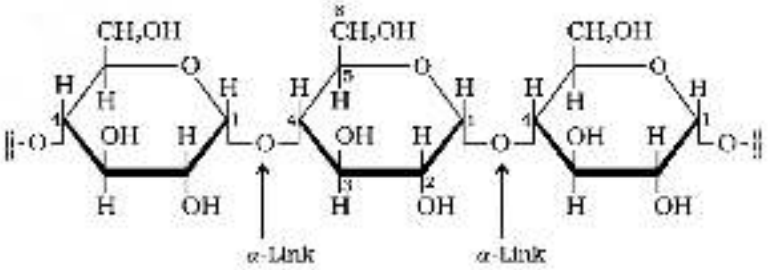
(1) Four

(2) Three

(3) Two

(4) One

67) The given diagram is of -



(1) Amylose

(2) Amylopectin

(3) Glycogen

(4) Both 1 & 3

68) Which protein is found in maximum amount in bio-sphere

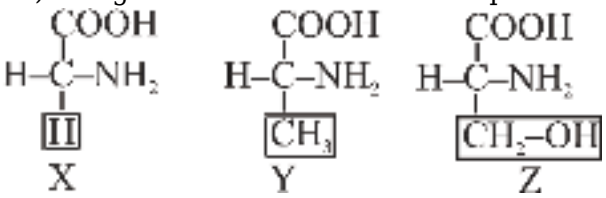
(1) Catalase

(2) Carbonic anhydrase

(3) Collagen

(4) RUBISCO

69) The given structural formulas represent amino acids (labelled by as X, Y and Z)



Identify the correct name of X, Y and Z.

(1) Alanine, serine and glycine respectively.

(2) Tyrosine, cysteine and glutamic acid respectively.

(3) Glycine, alanine and serine respectively.

(4) Cysteine, glutamic acid and tyrosine respectively.

70) Match the column I with column II and choose the correct answer :-

	Column I (Organic compound)		**Column II (Example)**
(A)	Fatty acid	(i)	Tryptophan
(B)	Complex polysaccharide	(ii)	Arachidonic acid
(C)	Amino acid	(iii)	RuBisCO
(D)	Protein	(iv)	Chitin

(1) A – ii, B – iii, C – i, D – iv

(2) A – ii, B – iv, C – i, D – iii

(3) A – iv, B – iii, C – ii, D – i

(4) A – ii, B – i, C – iii, D – iv

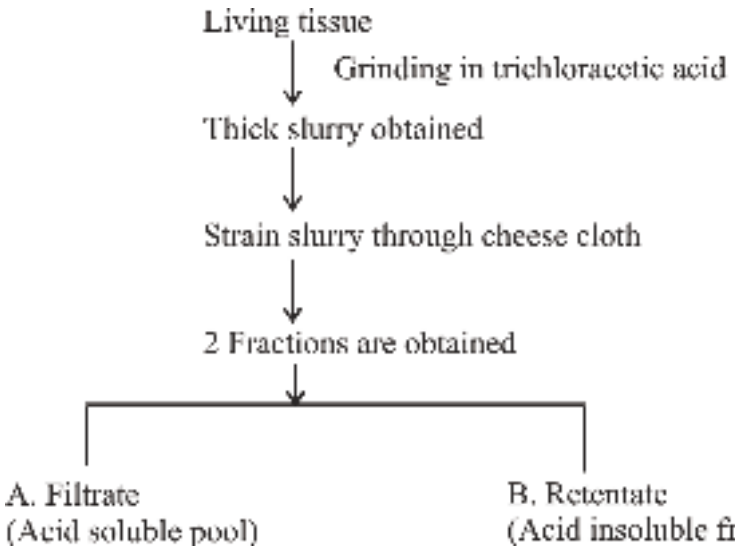
71) Correct order of abundances of O, S, Mg and Ca, in the earth crust ?

(1) O > Ca > Mg > S

(2) S > O > Mg > Ca

(3) Ca > O > S > Mg

(4) Mg > Ca > S > O



72) I. Molecular weight ranging from less than 1000 Dalton approximately
II. Has four type of organic compounds proteins, nucleic acids, polysaccharides and lipids
III. Contain chemicals that have molecular weight more than 1000 Dalton
IV. Has monomers
V. Has generally polymers except lipid
VI. Mostly micromolecules
VII. Generally macromolecules
How many of the above statements (I-VII) are correct regarding A & B respectively ?

	A	B
(1)	I, II, III	IV, V, VI, VIII
(2)	II, IV, VI	I, III, V, VII
(3)	I, IV, VI	II, III, V, VII
(4)	I, III, V	II, IV, VI, VII

(1) 1
(2) 2
(3) 3
(4) 4

73) Arachidonic acid has :-

(1) 20 carbon atoms including carboxyl carbon
(2) 18 carbon atoms including carboxyl carbon
(3) 16 carbon atoms including carboxyl carbon
(4) 22 carbon atoms including carboxyl carbon

74) How many of the following compounds are mono-saccharides :-
Glycerol, Lecithin, Cholesterol, Starch, Ribose, Triglyceride, Glucose, Mannose.

(1) 4
(2) 3
(3) 2

(4) 1

75) Choose the incorrect option for polysaccharides :-

(1) Cellulose is a homopolymer
(2) Inulin is a polymer of fructose
(3) Starch forms helical secondary structures
(4) Cellulose does contain complex helices and hence can hold iodine

76) Match the **column I** and **II** choose the correct option :-

	Column I		**Column II**
I	Trypsin	A	Hormone
II	Insulin	B	Fights infectious agent
III	Antibody	C	Enzyme
IV	GLUT-4	D	Enables glucose transport into cells

(1) I-A, II-B, III-C, IV-D
(2) I-C, II-A, III-B, IV-D
(3) I-B, II-C, III-A, IV-D
(4) I-D, II-A, III-D, IV-C

77)

Read the following and find out the correct statement

(1) Tertiary structure is not necessary for the many biological activities
(2) Monomeric protein show only quaternary structure
(3) In proteins, only right handed helices are observed
(4) In solution of different pHs, the structure of amino acid does not change.

78) Which of the following statements are not true ?
A. Glycerol is a 3-carbon compound with three–OH groups.
B. Waxes are monoesters with only one molecule of fatty acid attached with along chain monohydroxy alcohol.
C. Proteins are heteropolymers of nucleotides.
D. Stearic acid is a unsaturated fatty acid while oleic acid is saturated fatty acid.

(1) A & B only
(2) B & C only
(3) C & D only
(4) A & D only

79)

All the following statements about the cellulose are correct except :-

(1) Cellulose is a homopolymer

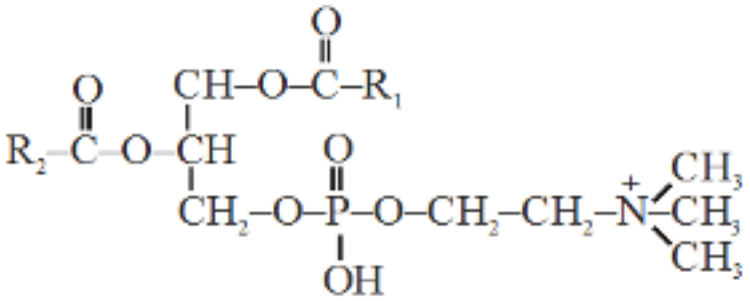
(2) It contain β(1-4) glycosidic linkage

(3) Cellulose is used for formation of paper

(4) Cellulose gives blue-black colour with iodine solution

80)

Identify the molecule given below with their respective statement :-



(1) Lecithin - Present in cell wall

(2) Cephalin - Present in blood

(3) Fatty acid - Present in ribosome

(4) Lecithin - Present in cell membrane

81) In a polypeptide or a protein, amino acids are linked by a peptide bond which is formed when the

(1) Carboxyl (–COOH) group of one amino acid reacts with the amino ($–NH_2$) group of the next amino acid with the addition of a water moiety

(2) Carboxyl (–COOH) group of one amino acid reacts with the amino ($–NH_2$) group of the next amino acid with the elimination of a water moiety

(3) Amino ($–NH_2$) group of one amino acid reacts with the R-group of the next amino acid with the elimination of a water moiety

(4) Amino ($–NH_2$) group of one amino acid reacts with the carboxyl (–COOH) group of the next amino acid with the addition of a water moiety

82) Which of the following statement is not true?

(1) Phospholipid is the most abundant lipid in cell membrane

(2) Cholesterol is a steroid

(3) Most common monomer of carbohydrate is glucose

(4) Galactose is a disaccharide form of carbohydrate

83) Find out non-essential fatty acid.

(1) Linolenic acid

(2) Arachidonic acids

(3) Linoleic acid

(4) Stearic acid

84) α-keratin protein assumes secondary structure through the formation of

(1) Intrachain hydrogen bond
(2) Interchain hydrogen bond
(3) Peptide bond
(4) Interchain disulphide bond

85) A protein that has three helical polypeptides supercoiled to form a rope like structure is

(1) Insulin
(2) Collagen
(3) Haemoglobin
(4) (1) and (3)

86) Read the following four statements (A-D) :-
(A) Dietary proteins are source of non-essential amino acids only.
(B) Protein are heteropolymer.
(C) Glycosidic bond and peptide bond are formed by dehydration.
(D) Choline is directly attached to glycerol by ester bond.
How many of the above statements are right ?

(1) Four
(2) Three
(3) Two
(4) One

87) Find the odd one out _____

(1) Cholesterol
(2) GLUT-4
(3) Receptor
(4) Trypsin

88)

Which of the following statement is wrong?

(1) Sucrose is a disacchride
(2) Cellulose is a polysacchride
(3) Uracil is pyrimidine
(4) Glycine is a sulpher containing amino acid

89) Units of cane sugar (CHINI) are:–

(1) α–D–Glucose + α – D–Glucose
(2) β–D–Glucose +β– D–Glucose

(3) α–D–Glucose + β – D–Fructose

(4) β–D–Glucose + α – D–Glucose

90) **Assertion :** When iodine comes into contact with starch, it forms a blue colour.
Reason : Starch have structure without helix , iodine trap in its branches to give blue colour.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	1	4	4	3	4	1	1	3	2	4	3	3	1	3	1	4	4	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	3	3	1	2	2	1	3	2	4	2	2	4	3	2	3	2	2	3	4
Q.	41	42	43	44	45															
A.	2	3	2	4	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	2	4	1	4	4	3	4	1	2	1	1	4	3	1	3	4	2	3	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	3	1	3	4	4	4	2	1	3	4	1	3	1	2	4	2	2	3	3
Q.	86	87	88	89	90															
A.	3	4	2	4	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	4	2	2	3	3	1	1	3	4	1	4	3	4	3	4	3	3	2	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	1	3	3	1	3	4	2	4	2	2	4	4	1	1	2	1	4	3	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	3	1	4	3	4	3	3	1	4	2	4	3	2	3	3	4	2	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	3	2	1	1	3	1	4	3	2	1	3	1	2	4	2	3	3	4	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	4	4	1	2	3	1	4	3	3										

SOLUTIONS

PHYSICS

1) $\vec{a}_{avg} = \dfrac{10\hat{j} - 10\hat{i}}{10}$

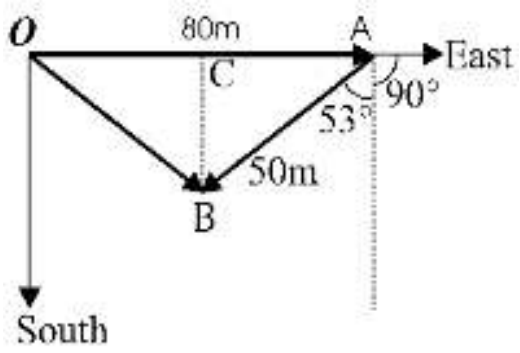
$\vec{a}_{avg} = \hat{j} - \hat{i}$

$\vec{a}_{avg} = \sqrt{2}\,m/s^2\ N - W$

2) $\vec{v}_{avg} = \dfrac{30\hat{i} + 40\hat{j}}{10}$

$|\vec{v}_{avg}| = 5m/s$

3)



AC = 40 m

CB = 30 m

OC = OA – AC = 40 m

$OB = \sqrt{OC^2 + CB^2}$ = 50 m

$\angle COB = 37^\circ$

$\overrightarrow{OB} = 50\ m$, 53° east of south

4)

In retardation $\vec{v}$ and $\vec{a}$ both are in opposite direction.

5) $V_{avg} = \dfrac{x}{\frac{2x}{5V_1} + \frac{3x}{5V_2}}$

$V_{avg} = \dfrac{5}{\frac{2}{V_1} + \frac{3}{V_2}} = \dfrac{5V_1V_2}{3V_1 + 2V_2}$

6)

mean speed = $\dfrac{\text{Total distance}}{\text{Total time}} = \dfrac{\pi r}{t_0}$

magnitude of mean velocity = $\frac{\text{displacement}}{\text{time}}$

$= \frac{2r}{t_0}$

7) $a = \frac{dv}{dt} = \frac{-1}{t^2} \Rightarrow \int_3^v dv = -\int_1^4 \frac{1}{t^2} dt$

$\Rightarrow \quad v - 3 = \left[\frac{1}{t}\right]^4 = \frac{1}{4} - 1$

$\Rightarrow \quad v = 3 - \frac{3}{4} = \frac{9}{4} = 2.25$ m/s.

8)

Slope of v-t graph = const.
a = const.

9) $S = \frac{1}{2}\left(\frac{\alpha\beta}{\alpha+\beta}\right)t^2$

t = 30 s

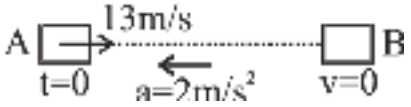
10) Average speed = 30/15 = 2 m/s

11) $h = \frac{1}{2}gt^2$

$h - \frac{7h}{16} = \frac{1}{2}g(t-1)^2$

$\frac{9}{16}\left(\frac{1}{2}gt^2\right) = \frac{1}{2}g(t-1)^2$

t = 4s

12) 

from A to B ⇒ v = u + at ⇒ 13 – 2t ⇒ t = 6.5 sec
It means velocity becomes zero at time 6.5 sec.
So distance travelled by body in 7th sec

$= 2 \times \frac{1}{2}at^2 \quad = 2 \times \frac{1}{2} \times 2 \times (0.5)^2 \; = 0.5$ m

13) $a = V\frac{dV}{dX}$

= (+ve) (-ve)
= –ve

14)

For a very short time interval, magnitude of instantaneous velocity will be equal to instantaneous speed.

15) $\frac{S_{4^{th}}}{S_{3^{rd}}} = \frac{\frac{a}{2}(7)}{\frac{a}{2}(5)} = \frac{7}{5}$

16)

$x = \sqrt{V+1}$

$x^2 = V + 1$

$V = x^2 - 1$

$a = V\frac{dV}{dx} = (x^2 - 1)(2x)$

at $x = 5$

$a = 24 \times 10 = 240$ unit

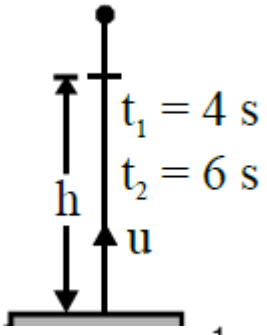
17) $x = (2t+5)^2 \Rightarrow v = 2(2t+5) \Rightarrow a = 4$

18) Velocity = slope of x– t graph.

19) At $t_1 = 4$ s and $t_2 = 6$ s, body is at height h

So $h = ut_1 - \left(\frac{1}{2}\right)gt_1^2$

and $h = ut_2 - \left(\frac{1}{2}\right)gt_2^2$



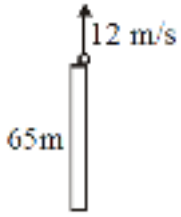
solve to get $u = \frac{1}{2}g(t_1 + t_2)$

$= \frac{1}{2} \times 10(4+6) = 50$ m/s

Alternate : Time of flight : $T = t_1 + t_2$

$\frac{2u}{g} = T \Rightarrow \frac{2u}{10} = 10 \Rightarrow u = 50$ m/s

20) b = –65 m
v = 12 m/s
a = –10 m/s^2



$-65 = 12t - \frac{1}{2} \times 10 \times t^2$
t = 5 sec

21)

$\because x = 2t^2 \quad \therefore v_x = \frac{dx}{dt} = 4t$

$\because y = t^2 - 4t \quad \therefore v_y = \frac{dy}{dt} = 2t - 4$

$\because z = 3t - 5 \quad \therefore v_z = \frac{dz}{dt} = 3$

For initial velocity : t = 0

$\therefore v = \sqrt{v_x^2 + v_y^2 + v_z^2} = \sqrt{(-4)^2 + (3)^2} = 5\,\text{unit}$

22)

$S_{n^{th}} = U + \frac{a}{2}(2n - 1)$

$S_{3^{rd}} = 0 + \frac{6}{2}(2 \times 3 - 1) = 15\text{m}$

23) $\because s = \frac{u^2}{2a} \quad \therefore \frac{s_1}{s_2} = \left(\frac{u_1}{u_2}\right)^2$

$\therefore \frac{90}{s_2} = \left(\frac{30}{50}\right)^2$

$\therefore s_2 = \frac{90 \times 2500}{900} = 250\text{m}$

24) $v_m = \sqrt{\frac{(7)^2 + (17)^2}{2}}$

$\therefore$ vm = 13 m/s

$\therefore x = \left(\frac{v_m + 7}{2}\right) t_1 = \left(\frac{13 + 7}{2}\right) t_1 = 10t_1$

& $x = \left(\frac{17 + v_m}{2}\right) t_2 = \left(\frac{17 + 13}{2}\right) t_2 = 15t_2$

$\therefore 10t_1 = 15t_2, \therefore \frac{t_1}{t_2} = \frac{3}{2}$

25) $\frac{S_{3rd}}{S_{2nd}} = \frac{10 + \frac{10}{2}(2 \times 3 - 1)}{10 + \frac{10}{2}(2 \times 2 - 1)} = \frac{7}{5}$

26)

$\Rightarrow S = Ut + \frac{1}{2}at^2$

$\Rightarrow 20 = \frac{U}{2} + \frac{1}{2}g\left(\frac{1}{2}\right)^2$

$\Rightarrow 20 = \frac{U}{2} + \frac{g}{8}$

U = 37.5 m/s

$\Rightarrow -h = U \times t - \frac{1}{2}gt^2$

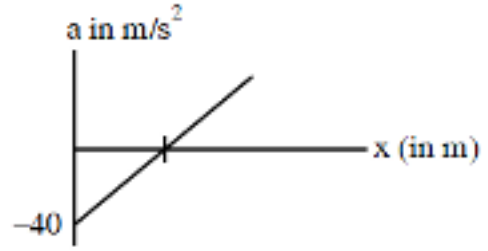
$\Rightarrow -20 = 37.5\ t - 5t^2$

$\Rightarrow t^2 - 7.5t - 4 = 0$

$\Rightarrow t = \frac{7.5 \pm \sqrt{(7.5)^2 - 4 \times 1 \times (-4)}}{2 \times 1}$

$\Rightarrow t = \frac{7.5 + 8.5}{2} = 8\ \text{sec}$

27) $V = -\frac{20}{10}x + 20$ $\Rightarrow V = -2x + 20$



$\Rightarrow \frac{dv}{dx} = -2$ $\Rightarrow a = v\frac{dv}{dx} = (-2x + 20)(-2)$

$\Rightarrow a = 4x - 40$

28)

$\Rightarrow v = 6t^2 + 2t + 3$

$\Rightarrow \frac{dS}{dt} = 6t^2 + 2t + 3$

$\Rightarrow \int_0^s dS = \int_0^1 \left(6t^2 + 2t + 3\right) dt$

$= 6\left[\frac{t^3}{3}\right]_0^1 + 2\left[\frac{t^2}{2}\right]_0^1 + 3\,[t]_0^1$

$\Rightarrow S = 2 + 1 + 3 = 6$ m

$V_{avg} = \frac{S}{t} = \frac{6}{1} = 6\,\text{m/s}$

29)

$\Rightarrow x = \frac{t^2}{2}$ $\Rightarrow v_x = \frac{2t}{2} = t$

$\Rightarrow y = \frac{x^2}{2}$ $\Rightarrow v_y = \frac{2x}{2} \times v_x = \frac{t^2}{2} \times t$

$\Rightarrow \vec{v} = v_x\hat{i} + v_y\hat{j} = t\hat{i} + \frac{t^3}{2}\hat{j}$

$\Rightarrow \vec{v}_{t=2} = 2\hat{i} + \frac{(2)^3}{2}\hat{j} = \left(2\hat{i} + 4\hat{j}\right)$ m/s

30) Maximum height = Area of V-t curve

$\frac{1}{2} \times 120 \times 1000 = 60\,\text{km}$

31) $a = v\left(\frac{dv}{dx}\right) = \left(3x^2 - 2x\right)(6x - 2)$

at $x = 2$, $a = 8 \times 10$

$a = 80\ m/s^2$

32) $x_P(t) = at + bt^2$ $x_Q(t) = ft - t^2$

$v_P = a + 2bt$ $v_Q = f - 2t$

as $v_P = v_Q$

$a + 2bt = f - 2t \Rightarrow t = \frac{f - a}{2(1 + b)}$

33) $\Delta v = 2v \sin\left(\frac{\theta}{2}\right)$

$= 2v \sin\left(\frac{60}{2}\right) = v$

34) **Question is Asking About:**

In this question we have to find the velocity with which the midpoint of the train passes the pole.

Given Data:

Acceleration of train = constant (say a)

Velocity of engine of train = u

Velocity of last compartment of train = v

Concept:

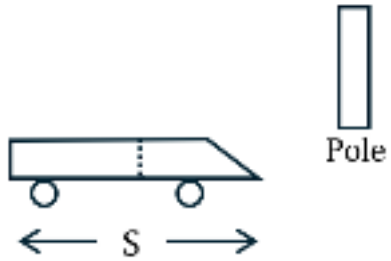
Equations of motion

Formula:

$v^2 = u^2 + 2as$

Solution/Explanation/Calculation:

Let the length of the train be 'S'



When the train completely passes the pole, the acceleration can be given by

$v^2 = u^2 + 2as$

or $\frac{v^2 - u^2}{2s} = a$ ..... (1)

Here u → velocity of engine of train (initial velocity)

v → velocity of last compartment of train (final velocity)

s → long in of train

Let the velocity of the midpoint of the train be v'

when passing in rough the pole.

Again applying equation of motion for midpoint

$(v')^2 = u^2 + 2a\,s'$

$\Rightarrow (v')^2 = u^2 + 2\left(\frac{v^2 - u^2}{2s}\right)\cdot\frac{s}{2}$ (here $s' = \frac{s}{2}$ for midpoint)

$\Rightarrow (v')^2 = \frac{u^2 + \frac{v^2 - u^2}{2}}{2}$

$\Rightarrow (v')^2 = \frac{\frac{u^2 - v^2}{2}}{2}$

$\Rightarrow v' = \sqrt{\frac{u^2 + v^2}{2}}$

Hence, option (3) is correct.

35) $\vec{u} = 4\hat{i} + 4\hat{j}$

$\vec{a} = -0.4\hat{j}$

$\vec{v} = \vec{u} + \vec{a}t$

$\vec{v} = (4 - 0.4t)\hat{i} + 4\hat{j}$

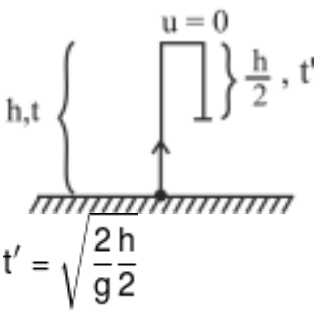
$|\vec{v}| = 5$

$\sqrt{(4 - 0.4t)^2 + 4^2} = 5$

4 – 0.4t = 3

t = 2.5 sec

36) $t = \sqrt{\frac{2h}{g}}$



$t' = \sqrt{\frac{2}{g}\frac{h}{2}}$

$$t' = \sqrt{\frac{h}{g}} = \frac{t}{\sqrt{2}}$$

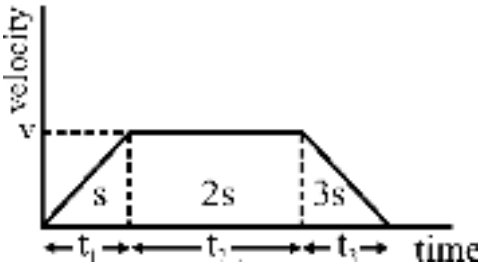
Total time $= t + t' = t + \frac{t}{\sqrt{2}}$

$$= t\left(1 + \frac{1}{\sqrt{2}}\right)$$

37) $$v_{avg} = \frac{6s}{t_1 + t_2 + t_3}$$

$$v_{avg} = \frac{6s}{\frac{2s}{v} + \frac{2s}{v} + \frac{6s}{v}}$$

$$v_{avg} = \frac{6s \times v}{10s} \Rightarrow \frac{v_{avg}}{v} = \frac{3}{5}$$



38)

$$\because \frac{da}{dt} = 2 \quad \therefore \int_0^a da = 2\int_0^2 dt$$

$$\int_0^a dV = 2\int_0^t tdt$$

$\therefore a = 2t$, $\therefore$, $v = t^2$

$$\int_0^s dS = \int_0^3 t^2 dt$$

$\therefore$, $\therefore S = \left[\frac{t^3}{3}\right]_0^3 = 9$ m

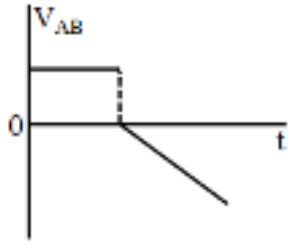
39) For motion, when body B moves from highest point to ground :-

$v_A = u_A - gt$ and $v_B = -gt$

$\therefore v_{AB} = v_A - v_B = u_A$ = const.

after body B remains on ground :

$v_{AB} = u_A - gt$ and $v_B = 0$



$\therefore v_{AB} = v_A - v_B = u_A - gt$

40)

 v = u + at v = (5 + 10) – 10 × 2
 v = 15 – 20 = –5 m/s
 v = 5 m/s

41) **Explanation :** A bus starts from rest and accelerates at 1 m/s^2. A man is 48 m behind the bus and moves with a uniform velocity of 10 m/s. We need to find the minimum time after which the man will catch the bus.

Concept : Relative Velocity and Relative Acceleration.

Formula : Relative Velocity $(\vec{v}_{rel}) = \vec{v}_{man} - \vec{v}_{bus}$
Relative Acceleration $(\vec{a}_{bus}) = \vec{a}_{man} - \vec{a}_{bus}$
Relative Displacement $(\vec{s}_{rel}) = \vec{s}_{man} - \vec{s}_{bus}$
$\vec{s}_{rel} = \vec{u}_{rel}t + (1/2)\vec{a}_{rel}t^2$

Calculation :
Relative Initial relative velocity (v_{rel}) = $v_{man} - v_{bus}$ = 10 – 0 = 10 m/s
Relative acceleration (a_{rel}) = $a_{man} - a_{bus}$ = 0 – 1 = –1 m/s^2
Relative displacement (s_{rel}) = 48 m (the initial distance between the man and the bus)

Apply the Equation of Motion for Relative Motion:
$s_{rel} = u_{rel}t + (1/2)a_{rel}t^2$
$48 = 10t + (1/2)(-1)t^2$
$48 = 10t - (1/2)t^2$
$96 = 20t - t^2$
$t^2 - 20t + 96 = 0$
(t – 12) (t – 8) = 0
t = 12 seconds or t = 8 seconds
the minimum time is the smaller of the two values, which is 8 seconds.
Hence, the correct answer is option (2).

42)

Say $\vec{a} = \hat{i}$
$\vec{u} = \hat{i} \times \left(\hat{i} \times \hat{i}\right) + \hat{j} \times \left(\hat{i} \times \hat{j}\right) + \hat{k} \times \left(\hat{i} \times \hat{k}\right)$
$\bar{u} = 0 + \hat{j} \times k + \hat{k} \times (-j)$
$\hat{j} \times \hat{k} + j \times \hat{k}$

$\bar{u} = 2\hat{i}\ \bar{u} = 2\vec{a}$

43) AB sinθ = $\sqrt{3}$AB cosθ
 tanθ = $\sqrt{3}$
θ = 60°
$\left|\hat{A} - \hat{B}\right| = 2a \sin\frac{\theta}{2} = 2a \sin 30° = a = 1$

$$V = -2x + 4$$

$$a = v\frac{dv}{dx}$$

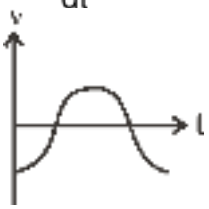
$$a = (-2x + 4) \times (-2)$$

44) $a = 4x - 8$

45)

$x = -\sin t$

$v = \frac{dx}{dt} = -\cos t$



CHEMISTRY

46)

According to the Bohr model of the atom, the energy required to remove an electron from a particular energy level to infinity is inversely proportional to the square of the principal quantum number (n) of that energy level. The energy required to remove an electron from the first energy level (n=1) to infinity is given as 200 units. The energy required to remove an electron from the second energy level (n=2) to infinity can be calculated using the formula:

$$E_n = \frac{E_1}{n^2}$$

where E_1 is the energy required to remove an electron from the first energy level to infinity, and n is the principal quantum number of the energy level from which the electron is being removed.

47)

Since $E_0 \propto me^4$, we can write $E' = \frac{1}{8} E_0$.

Substitute the value of E_0: $E' = \frac{1}{8}(-13.6\,\text{eV})$.

Calculate the new energy: $E' = -1.7\,\text{eV}$.

48) **Question Explanation:**
Which property increases when electron gets excited.

Concept:
This question is based on Bohr's atomic model.

Solution:
P.E. = $-27.2\frac{Z^2}{n^2}$ $n \uparrow$ P.E. $\uparrow$
$T = \frac{2\pi r}{V}$ $= T \alpha \frac{n^3}{Z^2}$ $n \uparrow$ $T \uparrow$
Orbit angular momentum $\Rightarrow mvr = \frac{nh}{2\pi}$ $n \uparrow$ Orbit angular momentum $\uparrow$

Final Answer: The correct option is (4).

49)

If two neutrons are added to an element, it will be converted into its isotope. Isotopes are variants of the same element that have the same number of protons (which defines the element) but different numbers of neutrons.

50)

The increasing order of e/m ratio (charge to mass ratio) for the particles, from lowest to highest, is: Neutron < Alpha particle < Proton < Electron.

51) $\frac{1}{\lambda} = R_H\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = R_H\left(\frac{1}{4} - \frac{1}{9}\right) = R_H \times \frac{5}{36}$
$\frac{1}{\lambda'} = R_H\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right) = \frac{R_H}{4}$
$\lambda - \lambda' = \frac{36}{5R_H} - \frac{4}{R_H} = \frac{1}{R_H}\left[\frac{36}{5} - 4\right] = 3.2/R_H$

52)

Option (3) is correct

53)

Radius of atom $r_a = 1A^0 = 10^{-10}m = 10^{-8}$ cm
Radius of Nucleus $r_N = 1.25 \times 10^{-13} \times A^{1/3}$ cm
$= 1.25 \times 10^{-13} \times (64)^{1/3}$ cm
$= 5.00 \times 10^{-13}$cm
$\frac{\text{Volume of Nucleus}}{\text{Volume of atom}} = \frac{\frac{4}{3}\pi r_N 3}{\frac{4}{3}\pi r_a 3} = \left(\frac{r_N}{ra}\right)^3$
$= \left(\frac{5 \times 10^{-13}}{10^{-8}}\right)^3$
$= 125 \times 10^{-15}$

$= 1.25 \times 10^{-13}$

54) % change in mass

$= \dfrac{\text{final mass} - \text{initial mass}}{\text{initial mass}} \times 100$

55)

Correct Option is (2)

$$r = \frac{n^2}{z} \times 0.529\text{Ã}...$$

$$4.768 = \frac{n^2}{1} \times 0.529$$

$$n = 3$$

$$\therefore \text{Angular momentum } (mvr) = \frac{nh}{2\pi} = \frac{3h}{2\pi}.$$

56)

$$\because r_n = 0.529 \times \left(\frac{n^2}{z}\right)\text{Å}$$

$\Rightarrow [r_n \alpha n^2]$

$\because$ Area $= \pi r^2$

57) $\therefore \left[\dfrac{A_1}{A_2} = \dfrac{r_1^2}{r_2^2} = \dfrac{n_1^4}{n_2^2}\right]$

$\because$ for 1st excited state means n = 2

$\therefore \dfrac{A_1}{A_2} = \dfrac{2^4}{1^4} = \dfrac{16}{1}$

58)

The wrong statement's are
(a) A and C - Isotones
(d) B and C - Isobars
(e) B and C - Isotopes

59)

A. Ionic mass = Number of protons + Number of neutrons.

B. Ionic mass =
Z+26=24+26=50Z plus 26 equals 24 plus 26 equals 50
Z+26=24+26=50

60)

Answer

61)

e^- jumps from higher to lower energy realises & lower to higher energy absorbed.

$$\frac{1}{\lambda} = R_H z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] \text{--}$$

$$\frac{c}{\lambda} = R_H c z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] \text{--}$$

$$\nu = R_H c z^2 \left[\frac{1}{1} - \frac{1}{4}\right] = \frac{3R_H c z^2}{4} \text{ [minimum]}$$

Ratio of the energy differences:

62) $$\frac{\Delta E_{21}}{\Delta E_{32}} = \frac{10.2}{\frac{17}{9}} = \frac{10.2 \times 9}{17} = \frac{91.8}{17} = \frac{27}{5}$$

63)

The two orbits are either I and II or II and IV

$\frac{r_{n_2}}{r_{n_1}} = \frac{4}{1}$ and $r_n \propto n^2$

Thus, $E_2 - E_1 = \frac{-13.6}{4} + 13.6 = 10.2$ eV

and $E_4 - E_2 = \frac{-13.6}{6} + \frac{-13.6}{4} = 2.55$ eV

64)

It is not true that cathode rays move at the same speed as light.

65) No. of deflected α-particles $\propto \frac{z^2}{\sin^4(\theta/2)}$

$$\frac{63}{x} = \frac{N(90^\circ)}{N(120^\circ)} = \frac{\sin^4(120/2)}{\sin^4(90/2)} = \frac{\sin^4(60^\circ)}{\sin^4(45^\circ)}$$

$$= \frac{\left(\frac{\sqrt{3}}{2}\right)^4}{\left(\frac{1}{\sqrt{2}}\right)^4} = \frac{\frac{9}{16}}{\frac{1}{4}} = \frac{9}{16} \times \frac{4}{1} = \frac{9}{4}$$

$$\frac{63}{x} = \frac{9}{4} \Rightarrow x = \frac{63 \times 4}{9} = 28$$

66)

The ratio of the time taken by the two electrons to complete one revolution is $\frac{1}{8}$

67)

Option (3) is correct

68)

Substitute the expression for v into the kinetic energy equation: KE = 1/2 * m * $(J/mr)^2$ = J^2 / $(2mr^2)$.

69) $r \propto \frac{n^2}{Z}$

70)

Option (4) is correct

71)

Option (4) is correct

72)

NCERT Page No. # 42

73) We know that when electrons of hydrogen atom return from higher energy levels such as 3, 4, 5, 6, 7 etc., to the lower energy level such as 2 or L-shell, a series of lines appear, which is called Balmer series.

$KE_{max} = h\nu - h\nu_0 \qquad ; V_0 = \frac{h\nu}{e} - \frac{h\nu_0}{e}$

74) slope 1 = h ; slope 2 = $\frac{h}{e}$

$$\frac{\text{slope1}}{\text{slope2}} = \frac{h}{h/e} = e$$

75)

When all wavelengths of radiation from ultraviolet to infrared are passed through hydrogen gas at room temperature, absorption lines will be observed in the Lyman series.

76)

Kinetic energy depends on frequnecy of incident radiation
$hV = hV_0 + 1/2mV^2$
with increase in intensity number of photo electrons increases

77)

For the shortest wavelength in the Lyman series of H (Z=1), this becomes:

$$\frac{1}{x} = R(1)^2\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right) = R \text{ or } x = \frac{1}{R}$$

Therefore, the longest wavelength in the Balmer series of He+ is

$$\frac{1}{\lambda_{max}} = R(2)^2\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = 4R\left(\frac{1}{4} - \frac{1}{9}\right) = 4R\left(\frac{5}{36}\right) = \frac{5R}{9}$$

$$\lambda_{max} = \frac{9}{5R} = \frac{9}{5}x$$

78)

Fact

79)

$$v_1 = R\,z^2\left(\frac{1}{(1)^2} - \frac{1}{(\infty)^2}\right) = R\ (z = 1)$$

$$v_2 = R\left(\frac{1}{(1)^2} - \frac{1}{(2)^2}\right) = \frac{3}{4}v_1 \quad v_3 = R = \left(\frac{1}{(1)^2} - \frac{1}{(\infty)^2}\right) = \frac{v_1}{4}$$

$v_1 - v_2 = v_3$

80) $E = \frac{12400}{3000} = 4.1\ eV$

K.E. = E – E_0
⇒ 0.5 = 4.13 – E_0 E_0 = 3.633 eV

81)

Fact

82) $2hf_0 = hf_0 + 1/2mv_1^2 \Rightarrow hf_0 = 1/2mv_1^2$
$5hf_0 = hf_0 + 1/2mv_2^2 \Rightarrow 4hf_0 = 1/2mv_2^2$
⇒ $\frac{1}{4} = \frac{v_1^2}{v_2^2} \Rightarrow v_1 : v_2 = 1 : 2$

83)

When an electron in the 10th energy level returns to the ground level (n=4), it can make the following transitions, all of which fall within the Brackett series:
10-4, 9-4, 8-4, 7-4, 6-4, and 5-4.

84)

Fact

85)

The energy change of 10.2 eV observed in the first Lyman transition of hydrogen is also observed in the second Balmer transition of the He+ ion. This occurs when an electron transitions from the n=4 to n=2 energy level in He+

86)

K.E. = h(ν – $ν_0$); x = h(ν – $ν_0$) .....(1)
$K.E._2$=h(2ν – $ν_0$) .....(2)

⇒ Divide equation (2) by (1)
⇒ $\frac{K.E._2}{x} = \frac{(2v - v_0)}{(v - v_0)}$
⇒ $\frac{K.E._2}{x} = \frac{v + v - v_0}{v - v_0}$
⇒ $\frac{K.E._2}{x} = \frac{v - v_0}{v - v_0} + \frac{v}{v - v_0}$
⇒ $\frac{K.E._2}{x} = 1 + y$
⇒ Since, y > 1. Hence $K.E._2$ > 2.

87) $2E - E - \frac{hc}{\lambda} \Rightarrow E - \frac{hc}{\lambda}$

$\frac{4E}{3} - E = \frac{hc}{\lambda'} \Rightarrow \frac{E}{3} = \frac{hc}{\lambda'} \therefore \frac{\lambda'}{\lambda} = 3 \Rightarrow \lambda' = 3\lambda$

88)

We are solving for n_2, the principal quantum number of the excited state. Let's rearrange the equation:

- $1/\lambda = R - R/n_2^2$
- $R/n_2^2 = R - 1/\lambda$
- $n_2^2 = \lambda R / (\lambda R - 1)$
- $n_2 = \sqrt{\lambda R} / \sqrt{(\lambda R - 1)}$

89)

The energy of the photon is the difference in energy levels: $E_{photon} = E_4 - E_2$.

Calculate E_4: $E_4 = -\frac{13.6}{4^2}\ \text{eV} = -\frac{13.6}{16}\ \text{eV} = -0.85\ \text{eV}$.

Calculate E_2: $E_2 = -\frac{13.6}{2^2}\ \text{eV} = -\frac{13.6}{4}\ \text{eV} = -3.4\ \text{eV}$.

Calculate E_{photon}: $E_{photon} = -0.85\ \text{eV} - (-3.4\ \text{eV}) = 2.55\ \text{eV}$.

The energy of the photon is the difference in energy levels: $E_{photon} = E_4 - E_2$.

Calculate E_4: $E_4 = -\frac{13.6}{4^2}\ \text{eV} = -\frac{13.6}{16}\ \text{eV} = -0.85\ \text{eV}$.

Calculate E_2: $E_2 = -\frac{13.6}{2^2}\ \text{eV} = -\frac{13.6}{4}\ \text{eV} = -3.4\ \text{eV}$.

Calculate E_{photon}: $E_{photon} = -0.85\ \text{eV} - (-3.4\ \text{eV}) = 2.55\ \text{eV}$.

90)

Fact

BIOLOGY

91)

NCERT Page no. 100

92) Module Pg. # 188

93)

Maximum difference in Length of two arms is in case of acrocentric chromosome

94)

Generated by Allie

Problem Statement: The question is asking which enzymatic protein found in the flagella has the ability to hydrolyze ATP, which means it can break down ATP molecules to release energy.

Underlying Concept: Flagella are cellular appendages that help in movement. Motor proteins like dynein in the flagella use the energy released from ATP hydrolysis to cause motion. Dynein is an ATPase enzyme that converts chemical energy from ATP into mechanical work, facilitating the bending movements of the flagella.

Tips and Tricks: Remember that dynein is the motor protein in flagella and cilia responsible for movement via ATP hydrolysis.

Common Mistakes: Confusing structural proteins like tubulin or actin with motor proteins; not recognizing the enzymatic role of dynein.

Why Other Options Are Incorrect?: Actin is mainly a cytoskeletal protein, Tubulin forms microtubules structurally but does not hydrolyze ATP for movement, and Nexin is a protein linking microtubules but not involved in ATP hydrolysis.

95) NCERT Pg.#140

96) NCERT XI, Pg. # 138

97) NCERT Pg. # 99

98)

Explanation: Euchromatin is genetically active because it is involved in transcription, while the others are genetically less active.
The correct answer is: 1. Euchromatin

99) NCERT (XI) Pg. # 137, para-8.5.8

100) NCERT Pg. # 138, Para-3, Line-24, 25, 26

101)

A genome is a haploid set of chromosomes. A haploid cell contains only one set of chromosomes, usually denoted as 'n'.

102) NCERT Pg. # 137, fig.-8.9 (b)

103) NCERT Pg # 139

104)

Arrangement of spindle fibers

Explanation :

A. The centrosome is an organelle in animal cells that plays a crucial role in organizing the spindle fibers during cell division (mitosis and meiosis). The centrosome contains two centrioles that help in the formation of the spindle apparatus, which is essential for the separation of chromosomes during cell division.

105)

These are smallest cell organelle is correct.
These are universal cell organelle is correct
Ribosomes are membrane less cell organelle is correct

106)

Important enzymes of peroxisome are catalase

107)

Organelle within organelle is Ribosome.

108)

5s mRNA is common to both 70 s & 80 s type of Ribosome

109)

Refer structure of cilia and fragella

110) NCERT-XI, Pg. # 99

111) NCERT XI, Page # 138, 8.5.10

112)

Refer topic centrosome and centriole

113)

Refer topic nucleus

114)

Generated by Allie

Problem Statement: The question asks which organelle inside a cell contains the enzymes necessary for the process of Î²-oxidation, which is the breakdown of fatty acids to generate energy.

Underlying Concept: Î²-oxidation is the metabolic process where fatty acids are broken down in cells to produce energy. This process requires specific enzymes found in certain organelles. Different organelles have distinct functions in a cell, so identifying the correct organelle that carries out Î²-oxidation is key. The peroxisome is known to contain enzymes for Î²-oxidation, particularly for very long-chain fatty acids.

Tips and Tricks: Remember that peroxisomes are involved in lipid metabolism; glyoxysomes also handle fatty acids but mainly in plant cells for converting fats into sugars.

Common Mistakes: Confusing peroxisomes with glyoxysomes or centrosomes might lead to an incorrect answer. Centrosomes are related to cell division, not metabolic processes.

Why Other Options Are Incorrect?: Glyoxysomes mainly occur in plant cells and are specialized peroxisomes involved in converting fats to carbohydrates but not primarily known for Î²-oxidation in animal cells. Centrosomes function as microtubule organizing centers and do not contain Î²-oxidation enzymes. Sphaerosomes are lipid storage structures and do not carry out Î²-oxidation.

115) NCERT-XI, Pg. # 139

116)

Refer topic lysosome

117)

Refer topic Nucleus

118)

Refer topic ribosome

119) **Solution/Explanation/Calculation:** Lamp brush chromosome: Discovered by Flamming and Ruckert from oocytes of vertebrates (Amphibia) during diplotene stage of cell division. Hence, option (4) is correct.

120)

Refer topic nucleus

121)

Refer topic nucleus

122)

Refer topic nucleus

123)

Refer topic endoplasmic Reticulum

124) NCERT (XIth) Pg. # 138 (8.5.10)

125) NCERT (XIth) Pg. # 138 (8.5.10)

126) NCERT-XI, Pg. # 100

127) NCERT (XIth) Pg. # 139

128) **The correct answer is : 4.** A-Nucleolus, B-Nuclear membrane, C-Nucleoplasm, D-Nuclear pore.



Explanation:

• Nucleolus (A): Dense region where r-RNA are made.
• Nuclear membrane (B): Double membrane surrounding the nucleus.

• Nucleoplasm (C): Fluid inside the nucleus.
• Nuclear pore (D): Openings in the nuclear membrane for transport.

129)

NCERT XI Pg # 100

130)

NCERT XI Pg.#138, 139

131) NCERT-XI, Pg. # 164, 165

132)

NCERT-XI, Pg. # 139, 140 (E + H)

133) NCERT Pg. # 127-129

134)

NCERT XI based

135)

Generated by Allie

Problem Statement: The question asks which structural feature is essentially present in a chromosome. The options include secondary constriction, centrioles, primary constriction, or both centrioles and primary constriction. The task is to identify the correct essential feature of chromosomes.

Underlying Concept: Chromosomes are thread-like structures composed of DNA and proteins, containing genes. A key structural feature of chromosomes is the constriction point called the centromere, which is also known as the primary constriction. This region is vital because it holds the two chromatids together and is the attachment site for spindle fibers during cell division. Centrioles, on the other hand, are cell organelles and not part of the chromosome structure. Secondary constrictions may be present in some chromosomes but are not an essential feature of all chromosomes.

Tips and Tricks: Remember that 'primary constriction' is synonymous with the centromere, a key part of every chromosome. Centrioles belong to the cell but are not on chromosomes.

Common Mistakes: Confusing centrioles (cell organelles) with centromeres (chromosome constriction). Also, secondary constrictions are not found on all chromosomes, so they are not essential.

Why Other Options Are Incorrect?: Secondary constriction is not present on all chromosomes and is not a defining feature. Centrioles are unrelated to chromosomes. The option of both centrioles and primary constriction is incorrect because centrioles are not part of chromosomes.

136)

NCERT Pg#108

137)

NCERT Pg#112

138)

NCERT Pg # 106

139)

140)

Generated by Allie

Problem Statement: The question asks us to identify which of the given molecular structures are zwitter ions. A zwitter ion is a molecule that contains both positive and negative charges at different locations within the same molecule.

Underlying Concept: Zwitter ions are molecules that have both a positively charged group and a negatively charged group, but the overall molecule is electrically neutral. In amino acids, this typically means having an 9NH3+ (ammonium) group and a 9COO- (carboxylate) group simultaneously.

Tips and Tricks: Focus on the presence of charged groups specifically 9NH3+ and 9COO- in the molecule to identify zwitter ions.

Common Mistakes: Confusing molecules that have only one charged group or lack either 9NH3+ or 9COO- as zwitter ions.

Why Other Options Are Incorrect?: Options without both the 9NH3+ and 9COO- groups are not zwitter ions and thus incorrect. For example, if a molecule has only a neutral or singly charged group, it is not zwitterionic.

141)

Question Explanation : Lipids - Acid insoluble fraction

Concept : This question is based on Lipids

Solution : Lipids form vesicles or micelle which are not water soluble.

Final Answer : option (2)

142)

Question Explanation : Insoluble fraction of tissue

Concept : This question is based on Tissue analysis - Chemical

Solution : Protein, Nucleic Acid, Polysaccharide,
Lipids → Acid insoluble fraction → Most of the Macromolecules

Final Answer : option (4)

143)

Question Explanation : Negligible amount - Human body

Concept : This question is based on Ash analysis

Solution : Silicon

Final Answer : option (3) Si

144)

NCERT Pg # 105

145)

NCERT Pg # 106

146)

NCERT Pg # 109

147) NCERT XI Pg.# 145

148) **Explanation:**
Oxygen makes up about 65% of our body mass, primarily as part of water (H_2O).**Therefore, the answer is Oxygen.**

The correct answer is 2.

149)

NCERT Pg # 110

150)

NCERT Pg # 106

151)

NCERT Pg # 105

152)

NCERT Pg # 112

153)

The Correct Answer 2.

The specificity of different types of proteins is primarily due to the **sequence of amino acids.**

Sequence of amino acids: This is the **correct** answer. The specific order of amino acids in a protein's polypeptide chain determines its three-dimensional structure, which in turn dictates its function.

154) NCERT Pg. # 148

155) **Question Explanation :**
Amino acid structure.

Concept :
This question is based on amino acid.

Solution :
Amino group and an acidic group as substituents on the same carbon:
This is correct. The acidic group refers to the carboxyl group (–COOH).

Hence,
Option (1) is correct

156)

NCERT Pg # 111

157)

NCERT Pg # 110

158)

Concept

A. RUBISCO (Ribulose-1,5-bisphosphate carboxylase/oxygenase): The enzyme responsible for carbon fixation in photosynthesis.

Explanation

A. RUBISCO is essential for photosynthesis, the process by which plants and algae convert carbon dioxide into organic compounds. Given the vast amount of photosynthetic organisms on Earth, RUBISCO is by far the most abundant protein.

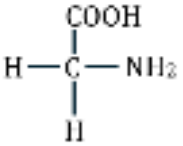
Final AnswerThe correct answer is (4) : RUBISCO.

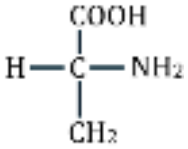
159)

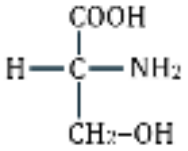
Question Explanation : Identification- x, y, z

Concept : This question is based on Amino acid

Solution :

x →  → Glycine

y →  → Alanine

z →  → Serine

Final Answer : option (3)

160) NCERT Pg # 111

161)

NCERT Pg # 105

162)

NCERT Pg # 105

163)

NCERT Pg # 106

164)

Concept:

A. Monosaccharides: Simple sugars that cannot be hydrolyzed into smaller carbohydrates. Examples include glucose, ribose, and mannose.

Explanation:

A. Ribose, Glucose and Mannose are monosaccharides.

Final Answer (2) : 3

165)

NCERT Pg # 110

166)

Correct answer : 2. I-C, II-A, III-B, IV-D

Explain : Trypsin is an enzyme, insulin is a hormone, antibodies fight infections, and GLUT-4 enables glucose transport.

167)

NCERT Pg # 112

168)

NCERT Pg # 106, 111

169)

NCERT Pg # 110

170)

NCERT Pg # 107

171) **Concept:**

A. Dehydration Synthesis: A reaction where a water molecule is removed to form a bond.

Explanation:

A. The carboxyl (-COOH) group of one amino acid reacts with the amino (-NH_2) group of another amino acid, releasing a water molecule (H_2O).
B. This creates a covalent bond (the peptide bond) between the carbon of the carboxyl group and the nitrogen of the amino group.
C. The R-groups are not involved in the formation of the peptide bonds.

Final Answer: 2. Carboxyl (-COOH) group of one amino acid reacts with the amino (-NH_2) group of the next amino acid with the elimination of a water moiety

172)

NCERT Pg # 106

173) **Concept :**
• **Non-essential Fatty Acids :** Fatty acids that can be synthesized by the body.
• **Stearic acid :** A saturated fatty acid that can be synthesized by the body.

Explanation :
• Stearic acid is a saturated fatty acid that the body can synthesize, making it a non-essential fatty acid.

174)

NCERT Pg # 112

175)

NCERT Pg # 112

176)

NCERT Pg # 107,111

177) NCERT [Pg-145,M]

178)

NCERT Pg # 106

179)

NCERT Pg # 110

180)

Concept:

A. Iodine and Starch Reaction: Iodine forms a blue color when it binds to starch, specifically to the helical structure of amylose (a component of starch).

B. The reason provided incorrectly describes the structure of starch, suggesting no helix and stating that iodine traps in branches, which is not accurate.

Explanation:

A. The Assertion is true: Iodine reacts with starch to give a blue color.

B. The Reason is incorrect: Starch has a helical structure in amylose, not without it. Iodine fits into this helix to form the blue color.